

## 12.5 Classwork: Roots of Complex Numbers — Markscheme

*No calculator. Marks are not printed on the student sheet; they are recorded here. Total: 26 marks.*

1. [Maximum mark: 6]

(a)  $(a + bi)^2 = a^2 - b^2 + 2abi = 5 - 12i$ , so  $a^2 - b^2 = 5$  and  $2ab = -12$  **M1**

Substituting  $b = -\frac{6}{a}$ :  $a^2 - \frac{36}{a^2} = 5$ ,  $a^4 - 5a^2 - 36 = 0$ ,  $(a^2 - 9)(a^2 + 4) = 0$

$a^2 = 9$  (since  $a$  is real), so  $a = \pm 3$ ,  $b = \mp 2$  **A1**

The square roots are  $3 - 2i$  and  $-3 + 2i$ . **A1**

*M1 for expanding and equating both real and imaginary parts. Accept the modulus shortcut  $a^2 + b^2 = |5 - 12i| = 13$ , which with  $a^2 - b^2 = 5$  gives  $a^2 = 9$ ,  $b^2 = 4$  directly. A1 for  $a = \pm 3$  with  $b = \mp 2$ ; A1 for both roots paired correctly. Pairing the signs wrongly ( $3 + 2i$ ) earns the M1 only —  $2ab = -12$  forces opposite signs. Accept  $\pm(3 - 2i)$ .*

(b)  $z = \frac{2 \pm \sqrt{4 - 4(-4 + 12i)}}{2} = \frac{2 \pm \sqrt{4(5 - 12i)}}{2} = 1 \pm (3 - 2i)$  **M1**

$z = 4 - 2i$  or  $z = -2 + 2i$  **A1**

*M1 for reducing the equation to a square root of  $5 - 12i$ , by the quadratic formula or by completing the square,  $(z - 1)^2 = 5 - 12i$ . A1 for both solutions. The “hence” means the root from (a) must be used; a candidate who restarts with  $z = x + yi$  and solves correctly earns full marks with the feedback note that (a) had already done the work. FT: award the A1 for  $1 \pm$  the candidate’s own square root from (a).*

(c) The coefficient  $-4 + 12i$  is not real. Non-real roots come in conjugate pairs only when every coefficient is real, so the theorem does not apply here. **R1**

*R1 requires the reason to name the non-real coefficient. “Because  $4 - 2i$  and  $-2 + 2i$  are not conjugates” only restates the question and earns nothing.*

2. [Maximum mark: 7]

$$(a) |w| = \sqrt{(-8)^2 + (8\sqrt{3})^2} = \sqrt{256} = 16 \quad \mathbf{A1}$$

$w$  is in the second quadrant with  $\tan \alpha = \sqrt{3}$ , so  $\arg w = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$   $\mathbf{A1}$

$$w = 16e^{2\pi i/3}$$

The marks are independent.  $\arg w = -\frac{\pi}{3}$  (from  $\arctan(-\sqrt{3})$  with no quadrant check)

loses the second A1. Accept  $16 \operatorname{cis} \frac{2\pi}{3}$ .

$$(b) z^4 = 16e^{i(2\pi/3+2k\pi)}, \text{ so } z = 2e^{i(\pi/6+k\pi/2)} \quad \mathbf{M1}$$

$$k = 0, 1: z = 2e^{i\pi/6}, 2e^{2\pi i/3} \quad \mathbf{A1}$$

$$k = -1, -2: z = 2e^{-i\pi/3}, 2e^{-5\pi i/6} \quad \mathbf{A1}$$

M1 for the modulus  $\sqrt[4]{16} = 2$  with the argument  $\frac{2\pi/3 + 2k\pi}{4}$ , or for one root followed by rotations of  $\frac{\pi}{2}$ . A1 for any two correct roots; A1 for all four with every argument in  $-\pi < \theta \leq \pi$ .  $2e^{7\pi i/6}$  and  $2e^{5\pi i/3}$  are outside the range and lose the final A1 unless converted. FT: award the marks from the candidate's own  $r$  and  $\theta$  in (a).

$$(c) 2e^{i\pi/6} = 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = \sqrt{3} + i \quad \mathbf{A1}$$

FT: award A1 for the candidate's own first-quadrant root from (b) correctly converted.

(d)  $-8 - 8\sqrt{3}i = \bar{w}$ , so the solutions are the conjugates of those in (b):

$$z = 2e^{-i\pi/6}, 2e^{-2\pi i/3}, 2e^{i\pi/3}, 2e^{5\pi i/6} \quad \mathbf{A1}$$

A1 for all four. Since  $\bar{z}^4 = \overline{z^4}$ , each conjugate root solves the conjugate equation. Accept solving  $z^4 = 16e^{-2\pi i/3}$  from the start. FT: award A1 for the conjugates of the candidate's own roots from (b).

3. [Maximum mark: 7]

(a)  $z^3 = 1 = e^{2k\pi i}$ , so  $z = e^{2k\pi i/3}$ ,  $k = 0, 1, -1$  **M1**

$z = 1, \quad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i$  **A1**

Accept the factor route:  $z^3 - 1 = (z - 1)(z^2 + z + 1) = 0$ , then the quadratic formula on  $z^2 + z + 1 = 0$ . A1 for all three in Cartesian form; answers left as  $e^{\pm 2\pi i/3}$  earn the M1 only.

(b)  $\left(\frac{z-1}{2}\right)^3 = 1$ , so  $z-1 = 2$ ,  $2\left(-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i\right) = -1 \pm \sqrt{3}i$  **M1**

$z = 3, \quad \sqrt{3}i, \quad -\sqrt{3}i$  **A1**

M1 for  $z-1 = 2\omega$  with  $\omega$  each cube root of unity from (a). A1 for all three solutions. Finding only  $z = 3$  earns nothing. FT: award the marks for  $1 + 2\omega$  with the candidate's own roots from (a).

(c)  $(z-1)^3 = z^3 - 3z^2 + 3z - 1$  **M1**

so  $(z-1)^3 = 8 \iff z^3 - 3z^2 + 3z - 9 = 0$  **AG**

The roots  $\pm\sqrt{3}i$  are a conjugate pair:  $(z - \sqrt{3}i)(z + \sqrt{3}i) = z^2 + 3$  **M1**

$z^3 - 3z^2 + 3z - 9 = (z-3)(z^2 + 3)$  **A1**

First M1 for the binomial expansion of  $(z-1)^3$  with coefficients 1, 3, 3, 1 and alternating signs. Second M1 for any valid route to the factors: pairing the conjugate roots from (b), grouping  $z^2(z-3) + 3(z-3)$ , or dividing by  $(z-3)$ . A1 for  $(z-3)(z^2 + 3)$ . A factorization with non-real coefficients, such as  $(z-3)(z - \sqrt{3}i)(z + \sqrt{3}i)$ , earns the second M1 only — the question asks for a real quadratic factor.

4. [Maximum mark: 6]

(a)  $1 + \sqrt{3}i = 2e^{i\pi/3}$

M1

$w = (2e^{i\pi/3})^3 = 8e^{i\pi} = -8$

A1

M1 for the exponential or polar form of  $1 + \sqrt{3}i$  (modulus 2, argument  $\frac{\pi}{3}$ ). A1 for  $w = -8$ . A correct binomial expansion of  $(1 + \sqrt{3}i)^3$  reaching  $-8$  also earns full marks, with the feedback note that the question asked for de Moivre.

(b) The cube roots of  $w$  are  $2e^{i\pi/3}$  rotated by  $\pm\frac{2\pi}{3}$ :

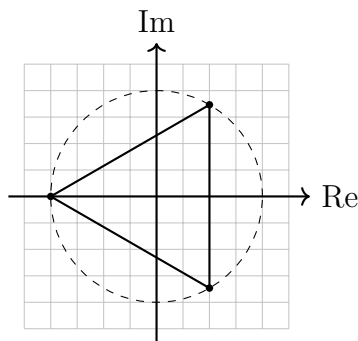
M1

$2e^{i\pi} = -2$  and  $2e^{-i\pi/3} = 1 - \sqrt{3}i$

A1

M1 for any valid method: rotation by  $\frac{2\pi}{3}$ , solving  $z^3 = 8e^{i(\pi+2k\pi)}$ , or the conjugate root theorem ( $z^3 + 8 = 0$  has real coefficients, so  $1 - \sqrt{3}i$  is a root) with the third root from the sum of roots  $= 0$  or from  $z = -2$  by inspection. A1 for both roots in Cartesian form. FT: award the marks for the cube roots of the candidate's own  $w$ .

(c)



The side from  $1 + \sqrt{3}i$  to  $1 - \sqrt{3}i$  has length  $2\sqrt{3}$ , and the height from  $-2$  to the line  $\text{Re}(z) = 1$  is 3.

M1

Area  $= \frac{1}{2} \times 2\sqrt{3} \times 3 = 3\sqrt{3}$

A1

M1 for any valid area method: base and height as above, the equilateral formula  $\frac{\sqrt{3}}{4}s^2$  with  $s = 2\sqrt{3}$ , or three triangles from the origin,  $3 \times \frac{1}{2}(2)(2) \sin \frac{2\pi}{3}$ . A1 for  $3\sqrt{3}$ . A decimal answer (5.20) loses the A1: the question asks for the exact area. FT: award the marks from the candidate's own roots in (b).